Maps API - Regional Data

Description

This request returns a cross section of regional data.

Enter a FRED series ID and your API key (optional) to get request parameters.

Request Wizard

Series ID *

Api Key

Generate

Parameters

- api_key
- file_type
- series_group
- region_type
- date
- start_date
- season
- units
- transformation
- frequency
- aggregation_method

Examples

This request can return either XML or JSON by setting the file_type parameter to xml or json. Note that the default value of file_type is xml. The API key 'abcdefghijklmnopqrstuvwxyz123456' is for demonstration purposes only. Use a registered API key instead.

XML

Request (HTTPS GET)

```
https://api.stlouisfed.org/geofred/regional/data?api_key=abcdefghijklmnopqrstuvwxyz123456&serie
```

Response

```
<?xml version="1.0" encoding="utf-8" ?>
    <series_data title="2013 Per Capita Personal Income by State (Dollars)" region="state" s
        <observation date="2013-01-01" region="Hawaii" code="15" value="43931" series_ic
        <observation date="2013-01-01" region="California" code="06" value="48502" series
        <observation date="2013-01-01" region="District of Columbia" code="11" value="67
        <observation date="2013-01-01" region="Colorado" code="08" value="47404" series_
        <observation date="2013-01-01" region="Connecticut" code="09" value="62647" ser
        <observation date="2013-01-01" region="West Virginia" code="54" value="35318" se
        <observation date="2013-01-01" region="Texas" code="48" value="44816" series_id=
        <observation date="2013-01-01" region="North Dakota" code="38" value="54900" se
        <observation date="2013-01-01" region="Wisconsin" code="55" value="43050" serie
        <observation date="2013-01-01" region="Washington" code="53" value="47857" seri
        <observation date="2013-01-01" region="New Hampshire" code="33" value="50865"
        <observation date="2013-01-01" region="Vermont" code="50" value="45770" series_
        <observation date="2013-01-01" region="New Jersey" code="34" value="54813" seri
        <observation date="2013-01-01" region="Virginia" code="51" value="48573" series_i
        <observation date="2013-01-01" region="New Mexico" code="35" value="35088" ser
        <observation date="2013-01-01" region="Tennessee" code="47" value="39454" serie
        <observation date="2013-01-01" region="Nevada" code="32" value="39664" series_i
        <observation date="2013-01-01" region="South Dakota" code="46" value="45608" se
        <observation date="2013-01-01" region="Mississippi" code="28" value="34259" seri
        <observation date="2013-01-01" region="New York" code="36" value="53395" series
        <observation date="2013-01-01" region="South Carolina" code="45" value="36204" s
        <observation date="2013-01-01" region="Missouri" code="29" value="40569" series_
        <observation date="2013-01-01" region="Ohio" code="39" value="41259" series_id="
        <observation date="2013-01-01" region="Minnesota" code="27" value="47838" serie
        <observation date="2013-01-01" region="Oklahoma" code="40" value="44017" serie
        <observation date="2013-01-01" region="Michigan" code="26" value="39566" series_
        <observation date="2013-01-01" region="Oregon" code="41" value="39788" series_i
        <observation date="2013-01-01" region="Maine" code="23" value="40329" series_id=
        <observation date="2013-01-01" region="Pennsylvania" code="42" value="46213" se
        <observation date="2013-01-01" region="Maryland" code="24" value="52249" series
        <observation date="2013-01-01" region="Wyoming" code="56" value="53776" series
        <observation date="2013-01-01" region="Louisiana" code="22" value="41376" series
        <observation date="2013-01-01" region="Massachusetts" code="25" value="56628"
        <observation date="2013-01-01" region="Kentucky" code="21" value="36058" series_
        <observation date="2013-01-01" region="North Carolina" code="37" value="38394" s
        <observation date="2013-01-01" region="Kansas" code="20" value="46174" series_i
        <observation date="2013-01-01" region="Nebraska" code="31" value="46653" series
        <observation date="2013-01-01" region="Indiana" code="18" value="39766" series_i
        <observation date="2013-01-01" region="Rhode Island" code="44" value="45447" se
        <observation date="2013-01-01" region="Illinois" code="17" value="47047" series_id
        <observation date="2013-01-01" region="Utah" code="49" value="36511" series_id="
        <observation date="2013-01-01" region="Idaho" code="16" value="36095" series_id=
        <observation date="2013-01-01" region="Alaska" code="02" value="52812" series_id
        <observation date="2013-01-01" region="Iowa" code="19" value="43439" series_id="
        <observation date="2013-01-01" region="Montana" code="30" value="40674" series_
```

```
        <observation date="2013-01-01" region="Alabama" code="01" value="36014" series_
        <observation date="2013-01-01" region="Georgia" code="13" value="37794" series_i
        <observation date="2013-01-01" region="Arkansas" code="05" value="36677" series
        <observation date="2013-01-01" region="Florida" code="12" value="41162" series_ic
        <observation date="2013-01-01" region="Arizona" code="04" value="37139" series_i
        <observation date="2013-01-01" region="Delaware" code="10" value="44404" series
    </series_data>
```


JSON

Request (HTTPS GET)

```
https://api.stlouisfed.org/geofred/regional/data?api_key=abcdefghijklmnopqrstuvwxyz123456&serie
```

Response

```
{
    "2013 Per Capita Personal Income by State (Dollars)": {
        "2013": [{
            "region": "Alabama",
            "code": "01",
            "value": "36481",
            "series_id": "ALPCPI"
        }, {
            "region": "Alaska",
            "code": "02",
            "value": "50150",
            "series_id": "AKPCPI"
        }, {
            "region": "Arizona",
            "code": "04",
            "value": "36983",
            "series_id": "AZPCPI"
        }, {
            "region": "Arkansas",
            "code": "05",
            "value": "36698",
            "series_id": "ARPCPI"
        }, {
            "region": "California",
            "code": "06",
            "value": "48434",
            "series_id": "CAPCPI"
        }, {
            "region": "Colorado",
            "code": "08",
            "value": "46897",
            "series_id": "COPCPI"
        }, {
            "region": "Connecticut",
            "code": "09",
            "value": "60658",
            "series_id": "CTPCPI"
        }, {
            "region": "Delaware",
            "code": "10",
            "value": "44815",
            "series_id": "DEPCPI"
        }, {
            "region": "District of Columbia",
            "code": "11",
            "value": "75329",
            "series_id": "DCPCPI"
```

```
        }, {
            "region": "Florida",
            "code": "12",
            "value": "41497",
            "series_id": "FLPCPI"
        }, {
            "region": "Georgia",
            "code": "13",
            "value": "37845",
            "series_id": "GAPCPI"
        }, {
            "region": "Hawaii",
            "code": "15",
            "value": "45204",
            "series_id": "HIPCPI"
        }, {
            "region": "Idaho",
            "code": "16",
            "value": "36146",
            "series_id": "IDPCPI"
        }, {
            "region": "Illinois",
            "code": "17",
            "value": "46980",
            "series_id": "ILPCPI"
        }, {
            "region": "Indiana",
            "code": "18",
            "value": "38622",
            "series_id": "INPCPI"
        }, {
            "region": "Iowa",
            "code": "19",
            "value": "44763",
            "series_id": "IAPCPI"
        }, {
            "region": "Kansas",
            "code": "20",
            "value": "44417",
            "series_id": "KSPCPI"
        }, {
            "region": "Kentucky",
            "code": "21",
            "value": "36214",
            "series_id": "KYPCPI"
        }, {
            "region": "Louisiana",
            "code": "22",
            "value": "41204",
            "series_id": "LAPCPI"
        }, {
```

```
            "region": "Maine",
            "code": "23",
            "value": "40924",
            "series_id": "MEPCPI"
        }, {
            "region": "Maryland",
            "code": "24",
            "value": "53826",
            "series_id": "MDPCPI"
        }, {
            "region": "Massachusetts",
            "code": "25",
            "value": "57248",
            "series_id": "MAPCPI"
        }, {
            "region": "Michigan",
            "code": "26",
            "value": "39055",
            "series_id": "MIPCPI"
        }, {
            "region": "Minnesota",
            "code": "27",
            "value": "47500",
            "series_id": "MNPCPI"
        }, {
            "region": "Mississippi",
            "code": "28",
            "value": "33913",
            "series_id": "MSPCPI"
        }, {
            "region": "Missouri",
            "code": "29",
            "value": "40663",
            "series_id": "MOPCPI"
        }, {
            "region": "Montana",
            "code": "30",
            "value": "39366",
            "series_id": "MTPCPI"
        }, {
            "region": "Nebraska",
            "code": "31",
            "value": "47157",
            "series_id": "NEPCPI"
        }, {
            "region": "Nevada",
            "code": "32",
            "value": "39235",
            "series_id": "NVPCPI"
        }, {
            "region": "New Hampshire",
```

```
            "code": "33",
            "value": "51013",
            "series_id": "NHPCPI"
        }, {
            "region": "New Jersey",
            "code": "34",
            "value": "55386",
            "series_id": "NJPCPI"
        }, {
            "region": "New Mexico",
            "code": "35",
            "value": "35965",
            "series_id": "NMPCPI"
        }, {
            "region": "New York",
            "code": "36",
            "value": "54462",
            "series_id": "NYPCPI"
        }, {
            "region": "North Carolina",
            "code": "37",
            "value": "38683",
            "series_id": "NCPCPI"
        }, {
            "region": "North Dakota",
            "code": "38",
            "value": "53182",
            "series_id": "NDPCPI"
        }, {
            "region": "Ohio",
            "code": "39",
            "value": "41049",
            "series_id": "OHPCPI"
        }, {
            "region": "Oklahoma",
            "code": "40",
            "value": "41861",
            "series_id": "OKPCPI"
        }, {
            "region": "Oregon",
            "code": "41",
            "value": "39848",
            "series_id": "ORPCPI"
        }, {
            "region": "Pennsylvania",
            "code": "42",
            "value": "46202",
            "series_id": "PAPCPI"
        }, {
            "region": "Rhode Island",
            "code": "44",
```

```
            "value": "46989",
            "series_id": "RIPCPI"
        }, {
            "region": "South Carolina",
            "code": "45",
            "value": "35831",
            "series_id": "SCPCPI"
        }, {
            "region": "South Dakota",
            "code": "46",
            "value": "46039",
            "series_id": "SDPCPI"
        }, {
            "region": "Tennessee",
            "code": "47",
            "value": "39558",
            "series_id": "TNPCPI"
        }, {
            "region": "Texas",
            "code": "48",
            "value": "43862",
            "series_id": "TXPCPI"
        }, {
            "region": "Utah",
            "code": "49",
            "value": "36640",
            "series_id": "UTPCPI"
        }, {
            "region": "Vermont",
            "code": "50",
            "value": "45483",
            "series_id": "VTPCPI"
        }, {
            "region": "Virginia",
            "code": "51",
            "value": "48838",
            "series_id": "VAPCPI"
        }, {
            "region": "Washington",
            "code": "53",
            "value": "47717",
            "series_id": "WAPCPI"
        }, {
            "region": "West Virginia",
            "code": "54",
            "value": "35533",
            "series_id": "WVPCPI"
        }, {
            "region": "Wisconsin",
            "code": "55",
            "value": "43244",
```

```
                "series_id": "WIPCPI"
            }, {
                "region": "Wyoming",
                "code": "56",
                "value": "52826",
                "series_id": "WYPCPI"
            }]
        }
    }
```


Parameters

api_key

Read API Keys for more information.

- 32 character alpha-numeric lowercase string, required

file_type

A key or file extension that indicates the type of file to send.

- One of the following values: 'xml', 'json'
  xml = Extensible Markup Language. The HTTP Content-Type is text/xml (xml is not available for county data).
  json = JavaScript Object Notation. The HTTP Content-Type is application/json.

series_group

The ID for a group of series found in FRED.

- string, required

region_type

The region you want to pull data for.

- string, required
- One of the following values:
  - bea
  - msa

- frb
- necta
- state
- country
- county
- censusregion

date

The date you want to pull a series group data from.

- YYYY-MM-DD formatted string, required

start_date

The start date you want to request series group data from. This allows you to pull a range of data

- YYYY-MM-DD formatted string, optional

season

The seasonality of the series group.

- string, required, five possible SA (Seasonally Adjusted), NSA (Not Seasonally Adjusted), SSA (Smoothed Seasonally Adjusted), SAAR (Seasonally Adjusted Annual Rate), or NSAAR (Not Seasonally Adjusted Annual Rate)

units

The units of the series you want to pull.

- string, required

transformation

A key that indicates a data value transformation.

- string, optional, default: lin (No transformation)
- One of the following values: 'lin', 'chg', 'ch1', 'pch', 'pc1', 'pca', 'cch', 'cca', 'log'
  lin = Levels (No transformation)
  chg = Change
  ch1 = Change from Year Ago
  pch = Percent Change
  pc1 = Percent Change from Year Ago
  pca = Compounded Annual Rate of Change

cch = Continuously Compounded Rate of Change
cca = Continuously Compounded Annual Rate of Change
log = Natural Log

- For unit transformation formulas, see: https://alfred.stlouisfed.org/help#growth_formulas

frequency

Frequency automatically assigns the default frequency of the map when using the Request Wizard above. The parameter can be used as a frequency aggregation feature. The maps frequency aggregation feature converts higher frequency data series into lower frequency data series (e.g. converts a monthly data series into an annual data series). In maps, the highest frequency data is daily, and the lowest frequency data is annual. There are 3 aggregation methods available- average, sum, and end of period. See the aggregation_method parameter.

- string
- One of the following values: 'd', 'w', 'bw', 'm', 'q', 'sa', 'a', 'wef', 'weth', 'wew', 'wetu', 'wem', 'wesu', 'wesa', 'bwew', 'bwem'

Frequencies without period descriptions:

d = Daily
w = Weekly
bw = Biweekly
m = Monthly
q = Quarterly
sa = Semiannual
a = Annual

Frequencies with period descriptions:

wef = Weekly, Ending Friday
weth = Weekly, Ending Thursday
wew = Weekly, Ending Wednesday
wetu = Weekly, Ending Tuesday
wem = Weekly, Ending Monday
wesu = Weekly, Ending Sunday
wesa = Weekly, Ending Saturday
bwew = Biweekly, Ending Wednesday
bwem = Biweekly, Ending Monday

aggregation_method

- string, optional, default: avg
- One of the following values: 'avg', 'sum', 'eop'

avg = Average
sum = Sum
eop = End of Period

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